

Volume II · Chapter 7 — The Hugging Face Ecosystem · Labs

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-7-context.md` for the AI interaction log.

Prerequisites: Ch 6.

Learning outcomes — the student can: navigate the Hub; use transformers pipelines and AutoModel/AutoTokenizer; load and process data with datasets; run efficient inference; manage models and licenses locally.

Labs (hands-on) — 12 h:

#	Lab	h
7a	Load a biomedical model via pipeline; run NER/classification	3
7b	Load & preprocess a medical dataset with datasets (map/tokenize)	3
7c	Batch inference over a clinical corpus; save outputs	3
7d	Build a small reproducible inference script + pinned environment	3

Datasets/tools: HF transformers, datasets, evaluate; PubMedBERT and similar.

Assessment: inference-pipeline deliverable (**50%**); dataset-processing lab (**30%**); quiz (**20%**).

Key decisions: model choice/license; precision vs. memory; streaming vs. in-memory datasets.

References: plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory **8** + Lab **12** = **20**.